PREDICTIVE ANALYTICS

PROJECT REPORT

Bank Marketing Subscription Prediction Using Machine Learning

Submitted by

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Programme and Section: B. Tech CSE / K2P24QR

Course Code: INT234

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CERTIFICATE

This is to certify that Alok Kumar bearing Registration no. 12414348 has completed project titled, " Bank Marketing Subscription Prediction Using Machine Learning" under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

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Date: October 30, 2026

DECLARATION

I, Alok Kumar, student of B.Tech Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 30/10/2026

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of my Data Science Minor Project titled **"Bank Marketing Subscription Prediction Using Machine Learning."**

First, I would like to thank my respected faculty guide, Dr. Mrinalini Rana, for providing valuable guidance, encouragement, and support throughout this project. Their suggestions and feedback helped me understand the concepts of Data Science and Machine Learning and complete this project successfully.

I would like to express my gratitude to all my teachers for sharing their knowledge and helping me develop my technical skills in Python, Data Analysis, Data Visualization, and Machine Learning.

I am also grateful to my friends and classmates for their cooperation, suggestions, and encouragement during the development of this project.

Finally, I would like to thank my parents and family members for their continuous support, motivation, and encouragement throughout my academic journey.

This project has been a valuable learning experience that enhanced my understanding of real-world data analysis and the practical application of machine learning algorithms.

ABSTRACT

The banking industry uses marketing campaigns to promote financial products and attract customers. However, not every customer responds positively to these campaigns. Identifying the customers who are more likely to subscribe can help banks understand customer behavior and improve their marketing strategies.

The main purpose of this project, **"Bank Marketing Subscription Prediction Using Machine Learning,"** is to analyze customer data and develop a machine learning classification model to predict whether a customer will subscribe to a bank's term deposit.

The project uses a bank marketing dataset containing 41,188 records and 21 attributes before duplicate removal. After data cleaning, 41,176 records were retained. The dataset contains customer information, campaign details, contact methods, and economic indicators. The target variable represents whether a customer subscribed to the term deposit.

Data preprocessing was performed using Python and Pandas. Exploratory Data Analysis (EDA) was conducted to investigate six objectives, including overall subscription rate, age group analysis, call duration, job and education, previous campaign outcomes, and contact methods. Various charts and visualizations were created using Matplotlib and Seaborn.

Four machine learning algorithms were implemented: Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and Naive Bayes. The dataset was divided into training and testing sets, and model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

Among the four models, Naive Bayes achieved the highest F1-score of 0.380 and recall of 72.30%, while Decision Tree achieved the highest accuracy of 89.83%. Naive Bayes was selected as the best-performing model based on the F1-score criterion.

The project demonstrates how data analysis and machine learning techniques can help understand customer subscription behavior and support data-driven marketing decisions in the banking sector.

Keywords: Data Science, Machine Learning, Bank Marketing, Customer Subscription, Classification, Python, Exploratory Data Analysis, Logistic Regression, KNN, Decision Tree, Naive Bayes.

1. INTRODUCTION

1.1 Background of the Project

In today's digital world, banks and financial institutions use data analysis and technology to understand customer behavior and improve their services. Marketing campaigns are an important part of banking because they help banks promote financial products and attract new customers.

One of the common financial products offered by banks is a term deposit. A term deposit is a financial product in which customers deposit a fixed amount of money for a specific period at an agreed interest rate.

Banks conduct marketing campaigns through different communication channels, such as telephone calls and cellular contacts, to encourage customers to subscribe to these financial products.

However, not every customer responds positively to marketing campaigns. Some customers subscribe to the offered product, while others reject the offer. Therefore, understanding customer behavior and identifying the factors associated with subscription decisions is important for improving marketing strategies.

Data Science and Machine Learning provide useful techniques for analyzing large datasets, discovering patterns, and predicting possible customer decisions.

1.2 Problem Statement

Traditional marketing campaigns may involve contacting a large number of customers without knowing whether they are likely to subscribe to a financial product.

This can result in unnecessary marketing expenses, inefficient use of resources, and lower campaign effectiveness.

The main problem addressed in this project is to analyze customer information and develop a machine learning model that can predict whether a customer will subscribe to a bank's term deposit.

The project uses historical bank marketing data to identify patterns and evaluate different classification algorithms.

1.3 Aim of the Project

The main aim of this project is to analyze bank marketing data and develop a machine learning-based classification system that predicts customer subscription decisions.

The project also aims to understand the relationship between customer characteristics, previous campaign outcomes, contact methods, and subscription behavior.

1.4 Objectives of the Project

The following six objectives are considered in this project:

1. Subscription Rate Analysis: Calculate the overall percentage of customers who subscribed and those who did not subscribe to the bank's term deposit.

2. Age Group Analysis: Analyze customer age groups and identify which groups have higher subscription rates.
3. Call Duration Analysis: Examine the relationship between call duration and customer subscription decisions.
4. Job and Education Analysis: Compare subscription rates across different occupations and education levels.
5. Previous Campaign Analysis: Investigate how previous marketing campaign outcomes are associated with current subscription decisions.
6. Contact Method Analysis: Compare subscription rates between different communication methods, such as cellular and telephone.

In addition to these objectives, four machine learning classification algorithms are implemented and compared to identify the best-performing model.

1.5 Scope of the Project

This project focuses on analyzing historical bank marketing data and applying machine learning techniques to predict customer subscription outcomes.

The scope includes:

- Loading and understanding the bank marketing dataset.
- Checking and removing duplicate records.
- Checking missing values and data types.
- Performing exploratory data analysis.
- Creating meaningful visualizations.
- Analyzing customer characteristics and campaign information.
- Preparing data for machine learning.
- Implementing four classification algorithms.
- Evaluating model performance using suitable metrics.
- Comparing models and selecting the best-performing model.
- Identifying important features using Decision Tree feature importance.

The project is developed using Python in the Jupyter Notebook environment.

1.6 Tools and Technologies Used

The following tools and technologies are used in the project:

1. Python: Python is the main programming language used for data preprocessing, analysis, visualization, and machine learning implementation.

2. Jupyter Notebook: It provides an interactive environment for writing and executing Python code and displaying results.

3. Pandas: Used for loading, cleaning, manipulating, and analyzing the dataset.

4. NumPy: Used for numerical operations and calculations.

5. Matplotlib: Used to create graphs and visualizations.

6. Seaborn: Used to create statistical visualizations and heatmaps.

7. Scikit-learn: Used for data preprocessing, feature scaling, model training, prediction, and model evaluation.

8. Microsoft Excel: Used to store and view the dataset.

1.7 Expected Outcome

The expected outcome of this project is to develop an understandable machine learning classification system that can identify patterns in bank marketing data and predict customer subscription decisions.

The project also aims to compare different classification algorithms and understand their strengths and limitations using performance metrics such as accuracy, precision, recall, and F1-score.

The final analysis can provide useful insights into customer behavior and help financial institutions understand how data-driven approaches can support marketing decisions.

2. DATASET DESCRIPTION AND SOURCE

2.1 Introduction to the Dataset

The dataset used in this project is a Bank Marketing dataset containing information about customers contacted during bank marketing campaigns. The dataset is used to analyze customer responses and understand the factors associated with subscribing to a term deposit.

It contains customer demographic information, financial campaign details, previous campaign outcomes, communication methods, and economic indicators.

The dataset provides information that can be used to identify patterns in customer subscription behavior and develop machine learning classification models.

The dataset was provided in Microsoft Excel format and loaded into Python using the Pandas library.

2.2 Dataset Information

The original dataset contains 41,188 records and 21 attributes.

The important dataset details are:

- Dataset name: Bank Marketing Dataset
- File format: Excel (.xlsx)
- Original number of records: 41,188
- Number of attributes: 21
- Duplicate records identified: 12
- Records after duplicate removal: 41,176

- Missing values: 0
- Target variable: y
- Target categories: yes and no

2.3 Source of the Dataset

The dataset was obtained as a bank marketing subscription dataset and stored locally in Microsoft Excel format under the filename "bank marketing subscription.xlsx".

The dataset is used for educational and analytical purposes to understand customer subscription behavior and implement machine learning classification algorithms.

The exact download platform and original dataset URL should be added here based on the source from which the dataset was downloaded.

2.4 Description of the Target Variable

The target variable used in this project is y.

It represents whether a customer subscribed to the bank's term deposit.

- Yes: The customer subscribed to the term deposit.
- No: The customer did not subscribe to the term deposit.

This variable is used as the output label for all four machine learning classification models.

2.5 Dataset attributes

Attribute	Description
age	Age of the customer
job	Occupation of the customer
marital	Marital status
education	Education level
default	Credit default status
housing	Housing loan status
loan	Personal loan status
contact	Communication method
month	Month of contact
day_of_week	Day of contact
duration	Duration of the last contact
campaign	Number of contacts during the current campaign
pdays	Days since previous campaign contact; 999 indicates no previous contact
previous	Number of contacts before the current campaign
poutcome	Previous campaign outcome

emp.var.rate	Employment variation rate
cons.price.idx	Consumer price index
cons.conf.idx	Consumer confidence index
euribor3m	Three-month Euribor rate
nr.employed	Number of employees
y	Term-deposit subscription outcome

2.6 Dataset Source and Original Background

The Bank Marketing dataset used in this project was downloaded from Kaggle, an online platform that provides datasets for data analysis, machine learning, and research.

The dataset is based on direct marketing campaigns conducted by a Portuguese banking institution. The bank contacted customers through telephone calls to encourage them to subscribe to a term deposit. A term deposit is a financial product in which a customer deposits money with a bank for a fixed period under agreed conditions.

The dataset is associated with the Bank Marketing research published by Sérgio Moro, Paulo Cortez, and Paulo Rita. The original dataset is also available through the UCI Machine Learning Repository.

The dataset contains information about customers, their financial and personal characteristics, details of marketing contacts, previous campaign outcomes, and economic indicators.

2.7 Dataset Structure

The dataset contains 41,188 rows and 21 columns before duplicate removal. Each row represents a customer contact record, while each column represents a particular customer attribute, campaign detail, economic indicator, or subscription outcome.

The dataset can be divided into the following major groups:

1. Customer Personal Information

These attributes describe the basic characteristics of customers.

- age: Age of the customer.
- job: Type of occupation.
- marital: Marital status.
- education: Education qualification.

2. Customer Financial Information

These attributes provide information about the customer's financial situation.

- default: Whether the customer has credit in default.
- housing: Whether the customer has a housing loan.
- loan: Whether the customer has a personal loan.

3. Current Marketing Campaign Information

These attributes describe how the bank contacted the customer during the current campaign.

- contact: Communication method used.

- month: Month of the last contact.
- day_of_week: Day of the last contact.
- duration: Duration of the last contact in seconds.
- campaign: Number of contacts made during the current campaign.

4. Previous Campaign Information

These attributes provide information about earlier marketing interactions.

- pdays: Number of days since the customer was previously contacted. The value 999 indicates that the customer was not previously contacted.
- previous: Number of contacts made before the current campaign.
- poutcome: Outcome of the previous marketing campaign.

5. Economic and Social Indicators

These attributes represent economic conditions during the marketing period.

- emp.var.rate: Employment variation rate.
- cons.price.idx: Consumer price index.
- cons.conf.idx: Consumer confidence index.
- euribor3m: Three-month Euribor rate.
- nr.employed: Number of employees.

6. Target Variable

- y: Indicates whether the customer subscribed to the bank's term deposit. It contains two possible values: yes and no.

2.8 Target Variable Description

The target variable of this project is y. It is the output that the machine learning models are trained to predict.

The target variable contains two classes:

- **Yes:** The customer subscribed to the term deposit.
- **No:** The customer did not subscribe to the term deposit.

For machine learning, these two categories are converted into numerical values:

- No = 0
- Yes = 1

This conversion allows classification algorithms to process the target variable.

The main purpose of the project is to identify patterns in customer and campaign information that may help predict whether a customer will subscribe to a term deposit.

2.9 Dataset Size and Initial Information

The original dataset contains 41,188 records and 21 columns. After removing 12 duplicate records, 41,176 records remain for further analysis.

The dataset contains both numerical and categorical variables. Numerical variables include age, duration, campaign, and economic indicators. Categorical variables include job, marital status, education, contact method, and campaign outcome.

The dataset contains no missing values in the initial missing-value check. However, duplicate records were identified and removed during preprocessing.

2.10 Importance of the Dataset

The Bank Marketing dataset is useful for studying customer behaviour and marketing campaign performance. It provides information about customer characteristics, communication methods, previous campaign results, and economic conditions.

By analyzing these variables, the project aims to understand the factors associated with term deposit subscriptions. Machine learning models are then used to classify whether a customer is likely to subscribe.

The project also demonstrates the practical use of data science techniques such as data cleaning, exploratory data analysis, feature encoding, model training, and performance evaluation.

3.DATA PREPROCESSING AND CLEANING

3.1 Introduction

Data preprocessing is an important stage of a data science project. Raw datasets may contain duplicate records, missing values, incorrect data types, inconsistent values, and other data quality problems. These issues can affect the accuracy and reliability of analysis and machine learning models.

In this project, the Bank Marketing dataset was loaded into Python using the Pandas library. Initial inspection was performed to understand the structure, dimensions, data types, and quality of the dataset.

The preprocessing stage focused on checking missing values, identifying duplicate records, understanding variable types, and preparing the dataset for exploratory data analysis and machine learning.

3.2 Loading the Dataset

The dataset was loaded into a Jupyter Notebook using the Pandas library. Pandas provides functions for reading, organizing, filtering, and analyzing structured data.

The Excel file was imported using the read_excel() function.

Python code:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


df = pd.read_excel("bank marketing subscription.xlsx")
```

print(df.head())

The head() function displays the first five records of the dataset. This helps to understand the column names and the type of information stored in each column.

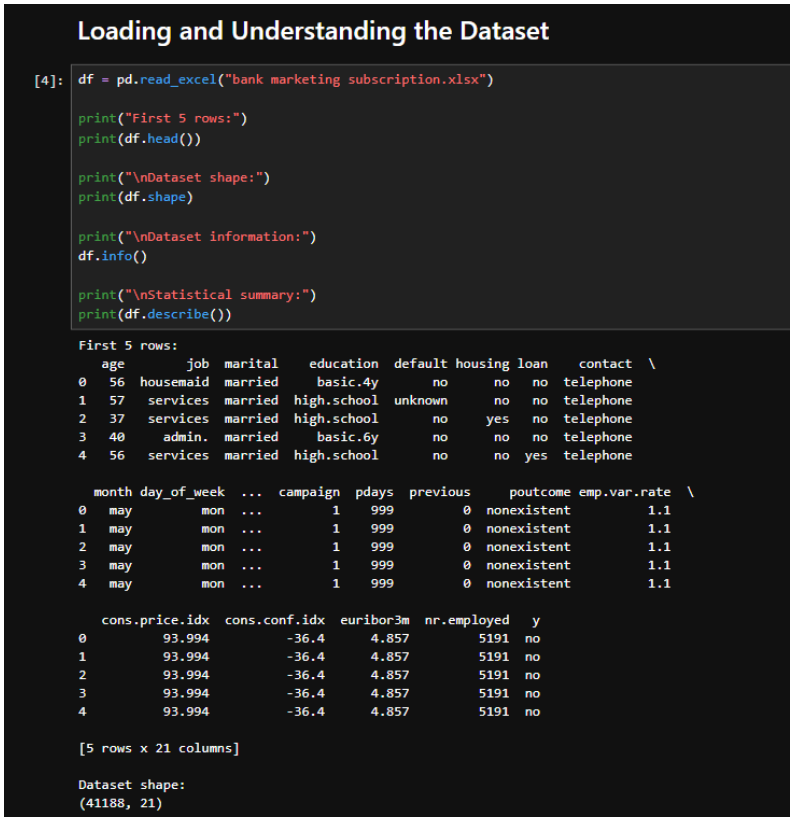


Figure 3.1: Loading and displaying the Bank Marketing dataset.

3.3 Checking Dataset Dimensions

The shape of the dataset was checked using the shape attribute.

Python code:

print(df.shape)

The original dataset contained 41,188 rows and 21 columns.

Rows represent customer contact records, while columns represent the different attributes associated with those records.



Figure 3.2: Original dataset dimensions.

3.4 Checking Dataset Information

The info() function was used to examine the structure of the dataset. It provides information about column names, non-null values, and data types.

Python code:

df.info()

The dataset contains numerical and categorical variables. Numerical variables store values such as age, campaign contacts, call duration, and economic indicators. Categorical variables store information such as occupation, education, marital status, and contact method.

This inspection helped identify the type of preprocessing required before applying machine learning algorithms.

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41188 entries, 0 to 41187
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   age             41188 non-null  int64
 1   job             41188 non-null  object
 2   marital         41188 non-null  object
 3   education       41188 non-null  object
 4   default         41188 non-null  object
 5   housing         41188 non-null  object
 6   loan            41188 non-null  object
 7   contact         41188 non-null  object
 8   month           41188 non-null  object
 9   day_of_week     41188 non-null  object
 10  duration        41188 non-null  int64
 11  campaign        41188 non-null  int64
 12  pdays           41188 non-null  int64
 13  previous        41188 non-null  int64
 14  poutcome        41188 non-null  object
 15  emp.var.rate    41188 non-null  float64
 16  cons.price.idx  41188 non-null  float64
 17  cons.conf.idx   41188 non-null  float64
 18  euribor3m       41188 non-null  float64
 19  nr.employed     41188 non-null  int64
 20  y               41188 non-null  object
dtypes: float64(4), int64(6), object(11)
memory usage: 6.6+ MB
```

Figure 3.3: Dataset information and data types.

3.5 Checking Missing Values

Missing values occur when information is not available for a particular record or attribute. Missing data can affect statistical analysis and machine learning performance.

The isnull().sum() function was used to count missing values in each column.

Python code:

```
print(df.isnull().sum())
```

The initial missing-value check showed that the dataset contained zero missing values across its 21 columns.

Therefore, no missing-value removal or imputation was required for the original dataset.

```
Missing values in each column:
age               0
job               0
marital           0
education         0
default           0
housing           0
loan              0
contact           0
month             0
day_of_week       0
duration          0
campaign          0
pdays             0
previous          0
poutcome          0
emp.var.rate      0
cons.price.idx    0
cons.conf.idx     0
euribor3m         0
nr.employed       0
y                 0
dtype: int64
```

Figure 3.4: Checking missing values in the dataset.

3.6 Checking Duplicate Records

Duplicate records are repeated rows that contain the same information. They can affect the results of data analysis by counting the same observation more than once.

The duplicated().sum() function was used to identify duplicate rows.

Python code:

```
print("Duplicate records:", df.duplicated().sum())
```

The initial inspection identified 12 duplicate records.

To remove these repeated records, the drop_duplicates() function was applied.

Python code:

```
df = df.drop_duplicates()


print("Shape after removing duplicates:", df.shape)
```

After removing the duplicate records, the dataset contained 41,176 rows and 21 columns.

This step helped ensure that repeated records were not included multiple times in the subsequent analysis.

```
Number of duplicate rows:
12

Column names:
['age', 'job', 'marital', 'education', 'default', 'housing', 'loan', 'contact', 'month', 'day_of_week', 'duration', 'campaign', 'pdays', 'previous', 'p
outcome', 'emp.var.rate', 'cons.price.idx', 'cons.conf.idx', 'euribor3m', 'nr.employed', 'y']
```

Figure 3.5: Removing duplicate records.

3.7 Checking Numerical Data

Numerical columns contain measurable values that can be analyzed using statistical methods.

The describe() function was used to calculate descriptive statistics such as:

- Count
- Mean
- Standard deviation
- Minimum value
- Maximum value
- 25th percentile
- 50th percentile
- 75th percentile

Python code:

```
df.describe()
```

These statistics help understand the distribution and range of numerical variables. They also provide an initial overview of customer characteristics and marketing campaign information.

```
Statistical summary:
                age      duration      campaign         pdays      previous  \
count  41188.00000  41188.000000  41188.000000  41188.000000  41188.000000
mean      40.02406    258.285010      2.567593    962.475454      0.172963
std       10.42125    259.279249      2.770014    186.910907      0.494901
min       17.00000      0.000000      1.000000      0.000000      0.000000
25%       32.00000    102.000000      1.000000    999.000000      0.000000
50%       38.00000    180.000000      2.000000    999.000000      0.000000
75%       47.00000    319.000000      3.000000    999.000000      0.000000
max       98.00000   4918.000000     56.000000    999.000000      7.000000

       emp.var.rate  cons.price.idx  cons.conf.idx     euribor3m   nr.employed
count  41188.000000    41188.000000   41188.000000  41188.000000  41188.000000
mean       0.081886       93.575664     -40.502600      3.621291   5167.019010
std        1.570960        0.578840       4.628198      1.734447     72.178074
min       -3.400000       92.201000     -50.800000      0.634000   4964.000000
25%       -1.800000       93.075000     -42.700000      1.344000   5099.000000
50%        1.100000       93.749000     -41.800000      4.857000   5191.000000
75%        1.400000       93.994000     -36.400000      4.961000   5228.000000
max        1.400000       94.767000     -26.900000      5.045000   5228.000000
```

Figure 3.6: Statistical summary of numerical variables.

3.8 Removing Duplicate Records

Removing duplicate records was the main cleaning operation required for this dataset.

The dataset originally contained 41,188 records. After identifying and removing 12 duplicate rows, 41,176 unique records remained.

Description	Number of records
Original records	41,188
Duplicate records removed	12
Final records	41,176
Total columns	21

3.9 Final Dataset After Preprocessing

After completing the initial data inspection and cleaning process, the dataset was ready for exploratory data analysis.

The final dataset contained 41,176 records and 21 columns. The initial missing-value check showed no missing values, and the identified duplicate records had been removed.

The cleaned dataset was stored in the DataFrame df and used for further analysis.

3.10 Conclusion

Data preprocessing helped improve the quality and consistency of the Bank Marketing dataset before performing exploratory data analysis and machine learning.

The initial inspection identified the dataset structure, numerical and categorical variables, and missing-value status. A total of 12 duplicate records were removed, reducing the dataset from 41,188 to 41,176 records.

The preprocessing stage provided a suitable foundation for the next stage of the project, which focuses on exploratory data analysis and understanding customer subscription patterns.

4.EXPLORATORY DATA ANALYSIS (EDA)

4.1 Introduction

Exploratory Data Analysis (EDA) is an important step in the data science process. It is performed to understand the structure, distribution, relationships, and patterns present in a dataset.

In this project, EDA was performed on the Bank Marketing dataset after the preprocessing and cleaning stage. The main purpose was to understand customer characteristics, marketing campaign activities, and the factors associated with term deposit subscriptions.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to perform statistical analysis and create visualizations.

EDA helped identify patterns in the dataset and provided a better understanding of the variables before building machine learning classification models.

4.2 Objectives of Exploratory Data Analysis

The main objectives of EDA in this project were:

1. Understand the distribution of the target variable.
2. Analyze subscription patterns across different customer age groups.
3. Examine the relationship between call duration and subscription.
4. Compare subscription rates across job and education categories.
5. Study the relationship between previous campaign outcomes and subscription.
6. Compare subscription rates across different contact methods.
7. Identify useful patterns and relationships among the dataset variables.

4.3 Tools and Libraries Used

The following Python libraries were used during exploratory data analysis.

Library	Purpose
Pandas	Data manipulation and analysis
NumPy	Numerical operations
Matplotlib	Creating graphs and charts
Seaborn	Statistical data visualization
Jupyter Notebook	Writing and executing Python code

Pandas: Pandas was used to read, organize, filter, group, and summarize the dataset.

NumPy: NumPy was used for numerical calculations and data operations.

Matplotlib: Matplotlib was used to create charts such as bar graphs and other visual representations.

Seaborn: Seaborn was used to create statistical visualizations and make relationships in the data easier to understand.

Jupyter Notebook: Jupyter Notebook provided an interactive environment for writing Python code, displaying outputs, and documenting the analysis.

4.4 Understanding Numerical Variables

Numerical variables contain measurable values that can be analyzed using statistical techniques.

In the Bank Marketing dataset, important numerical variables include:

- age
- duration
- campaign
- pdays
- previous
- emp.var.rate
- cons.price.idx
- cons.conf.idx
- euribor3m
- nr.employed

Descriptive statistics were used to understand the distribution and range of these variables.

The analysis of numerical variables helped identify customer age patterns, campaign contact frequency, call duration, and economic conditions.

4.5 Understanding Categorical Variables

Categorical variables represent groups or categories rather than continuous numerical measurements.

Important categorical variables in this dataset include:

- job
- marital
- education
- default
- housing
- loan
- contact
- month
- day_of_week
- poutcome
- y

These variables were analyzed using frequency counts, grouped summaries, and visualizations.

Categorical analysis helped compare customer subscription behaviour across different occupations, education levels, contact methods, and previous campaign outcomes.

4.6 Analysis of Target Variable Distribution

The target variable y represents whether a customer subscribed to a term deposit.

It contains two categories:

- Yes
- No

The distribution of the target variable was analyzed to understand the proportion of customers who subscribed and those who did not subscribe.

The analysis showed that:

- Subscription rate: 11.27%
- Non-subscription rate: 88.73%

This indicates that most customers in the dataset did not subscribe to the term deposit.

The target variable is therefore imbalanced because the number of non-subscribers is considerably higher than the number of subscribers.

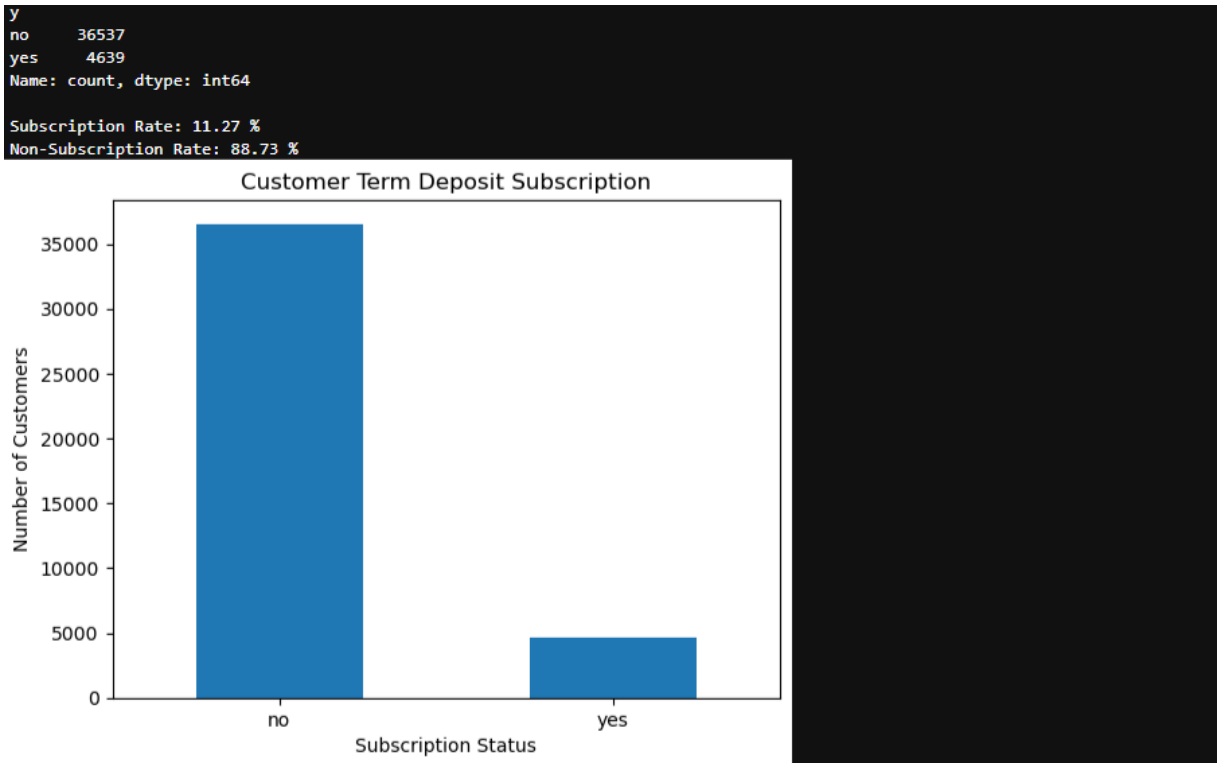


Figure 4.1: Distribution of customers who subscribed and did not subscribe.

4.7 Data Visualization Techniques

Data visualization is the graphical representation of information. It makes complex data easier to understand and helps identify patterns, comparisons, and relationships.

The following visualization techniques were used in this project:

1. Bar Chart

A bar chart represents categorical data using rectangular bars. It is useful for comparing subscription counts or rates across categories such as age groups, occupations, education levels, and contact methods.

2. Count Plot

A count plot displays the number of observations in different categories. It helps understand how frequently each category appears in the dataset.

3. Pie Chart

A pie chart represents the proportions of different categories as parts of a circle. It can be used to display the percentage of subscribers and non-subscribers.

4. Grouped Summary Tables

Grouped summaries calculate statistics for different categories. In this project, they were used to calculate subscription rates for different customer groups.

5. Statistical Summary

Descriptive statistics provide information such as mean, standard deviation, minimum, maximum, and quartiles for numerical variables.

4.8 Importance of EDA in This Project

EDA played an important role in understanding the Bank Marketing dataset before model development.

It helped identify the imbalance in the target variable and examine how subscription rates varied across customer groups and marketing campaign characteristics.

The analysis also helped determine which variables could provide useful information for the classification task.

For example, customer age, occupation, education, contact method, and previous campaign outcome were examined to understand their association with subscription behaviour.

The findings from EDA were used to guide the feature preparation and machine learning stages.

4.9 Summary of Exploratory Data Analysis

The exploratory data analysis provided an overall understanding of the dataset and its important characteristics.

The analysis identified that the subscription rate was 11.27%, while the non-subscription rate was 88.73%. This showed that the target variable was imbalanced.

Further analysis was performed to compare subscription behaviour across age groups, call durations, job categories, education levels, previous campaign outcomes, and contact methods.

The detailed results of these six objectives are presented in the next chapter.

4.10 Conclusion

Exploratory Data Analysis helped convert the cleaned dataset into meaningful information. Statistical summaries and visualizations were used to understand customer characteristics and marketing campaign patterns.

The analysis provided useful insights into the distribution of the target variable and the relationships between different customer and campaign attributes.

EDA also established a foundation for feature engineering and machine learning model development.

5.ANALYSIS OF PROJECT OBJECTIVES

5.1 Introduction

After completing data preprocessing and exploratory data analysis, a detailed analysis was performed to understand customer subscription behaviour in the Bank Marketing dataset.

The analysis was divided into six objectives. Each objective focuses on a different aspect of customer characteristics or marketing campaign activities.

The main purpose was to identify patterns and understand how different factors are associated with customers' decisions to subscribe to a term deposit.

The six objectives are:

1. Calculate the overall subscription rate.
2. Analyze subscription rates across different age groups.
3. Study the relationship between call duration and subscription.
4. Compare subscription rates by job and education.
5. Analyze the effect of previous campaign outcomes.
6. Compare subscription rates by contact method.

5.2 Objective 1: Calculate the Overall Subscription Rate

5.2.1 Objective

The first objective was to calculate the overall subscription rate in the Bank Marketing dataset.

The target variable y was used to identify whether customers subscribed to the term deposit or not.

The analysis helped determine the proportion of successful subscriptions compared with unsuccessful subscriptions.

5.2.2 Methodology

The total number of customers in each target category was calculated using the value_counts() function.

The percentage of each category was calculated by dividing its count by the total number of records and multiplying the result by 100.

Python code:

```
subscription_counts = df["y"].value_counts()


subscription_rate = (
    df["y"].value_counts(normalize=True) * 100
)


print(subscription_counts)
print(subscription_rate)
```

5.2.3 Analysis Results

The analysis produced the following results:

Subscription status	Percentage
Yes	11.27%
No	88.73%

The results show that only 11.27% of the customers subscribed to the term deposit, while 88.73% did not subscribe.

This indicates that the dataset contains a considerably higher number of non-subscribers than subscribers.

The difference between these two categories highlights the presence of class imbalance in the target variable.

5.2.4 Interpretation

The analysis indicates that successful subscriptions represent a relatively small proportion of the total customer records.

This finding is important for machine learning because a classification model may predict the majority class more frequently and still achieve high accuracy.

Therefore, accuracy alone may not provide a complete evaluation of model performance. Precision, recall, F1-score, and the confusion matrix are also important for evaluating the models.

5.2.5 Visualization

A bar chart was used to compare the percentage of customers who subscribed and those who did not subscribe.

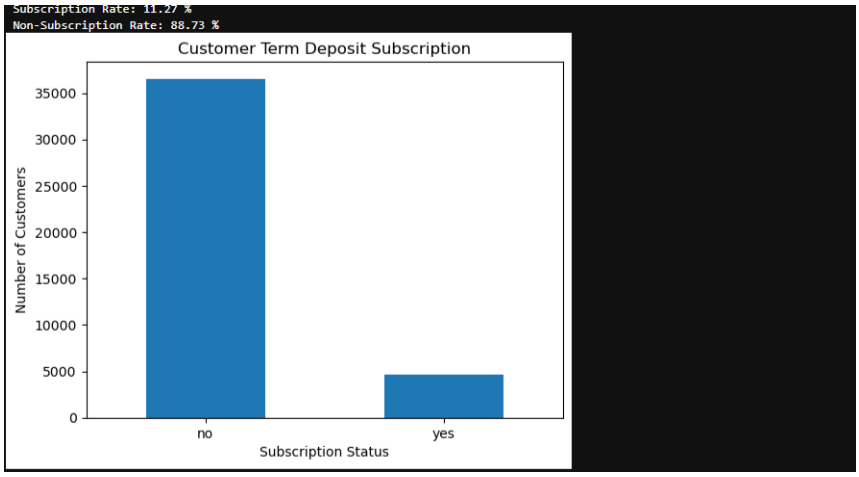


Figure 5.1: Overall Term Deposit Subscription Distribution.

5.3 Objective 2: Analyze Subscription Rates Across Different Age Groups

5.3.1 Objective

The second objective was to study how customer age is associated with term deposit subscription behaviour.

Age is an important customer characteristic that may help understand differences in customer preferences and responses to marketing campaigns.

For this analysis, customers were divided into different age groups, and the subscription rate of each group was calculated.

5.3.2 Methodology

A new categorical variable called age_group was created to divide customers into different age categories.

The age groups used in the analysis were:

- 18–25
- 26–35
- 36–45
- 46–55
- 56+

The subscription rate for each age group was then calculated.

Python code:

```
bins = [0, 25, 35, 45, 55, 100]

labels = [
    "18-25",
    "26-35",
    "36-45",
    "46-55",
    "56+"
]

df["age_group"] = pd.cut(
    df["age"],
    bins=bins,
    labels=labels
)

age_analysis = pd.crosstab(
    df["age_group"],
    df["y"],
    normalize="index"
) * 100
```

print(age_analysis)

Note: Check the minimum age in your dataset and confirm the age-group boundaries before final submission.

5.3.3 Analysis Results

The analysis produced the following subscription rates:

Age group	Subscription rate
18–25	20.95%
26–35	11.72%
36–45	8.51%
46–55	8.69%
56+	20.69%

5.3.4 Interpretation

The results indicate that the 18–25 age group had the highest subscription rate of 20.95%, followed closely by the 56+ age group at 20.69%.

The 36–45 age group recorded the lowest subscription rate at 8.51%.

The results suggest that subscription behaviour varies across different age groups. Younger and older customers showed higher subscription rates than the middle age groups in this analysis.

However, these findings represent associations within the dataset and do not establish that age directly causes a customer to subscribe.

5.3.5 Visualization

A bar chart was used to compare subscription rates across different age groups.

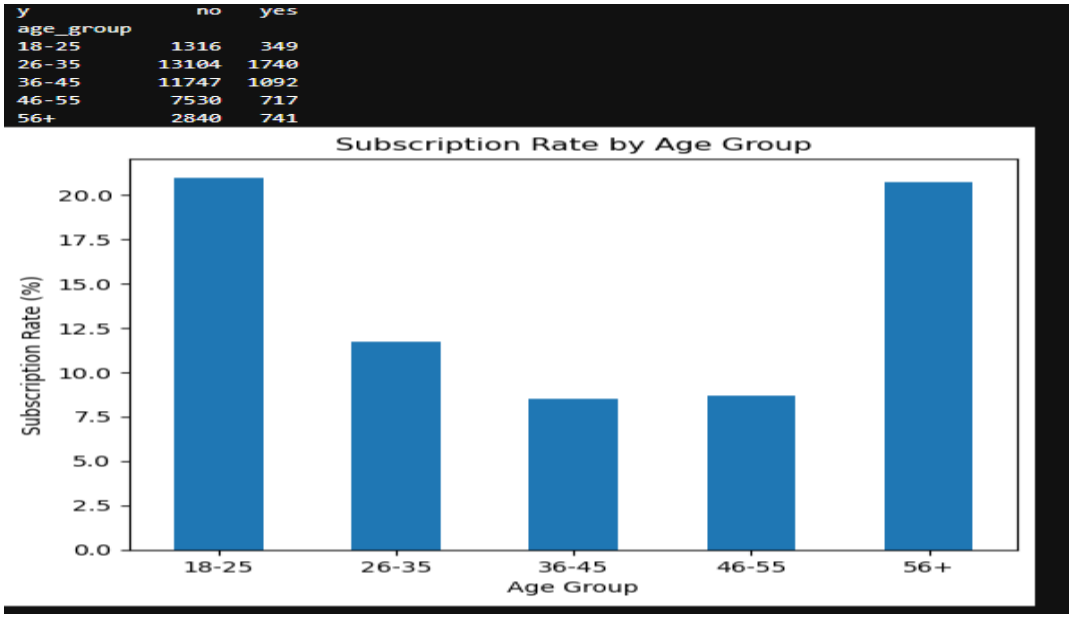


Figure 5.2: Term Deposit Subscription Rate by Age Group.

5.4 Objective 3: Study the Relationship Between Call Duration and Subscription

5.4.1 Objective

The third objective was to examine the relationship between the duration of a telephone call and the customer's subscription decision.

The duration column represents the duration of the last contact with the customer, measured in seconds.

This analysis helps understand whether customers who subscribed had different average call durations compared with customers who did not subscribe.

5.4.2 Methodology

The average call duration was calculated separately for customers who subscribed and those who did not subscribe.

The groupby() function was used to group the records according to the target variable.

Python code:

```
duration_analysis = df.groupby("y")["duration"].mean()


print(duration_analysis)
```

5.4.3 Analysis Results

The following average call durations were obtained:

Subscription status	Average duration
No	220.87 seconds
Yes	553.26 seconds

5.4.4 Interpretation

The results show that customers who subscribed had an average call duration of 553.26 seconds, while customers who did not subscribe had an average call duration of 220.87 seconds.

This indicates that successful subscription calls were considerably longer on average than unsuccessful calls.

One possible explanation is that longer conversations may provide more time for the bank representative to explain the term deposit and answer customer questions.

However, longer calls do not necessarily cause subscriptions. The relationship is an observed association.

An important consideration is that call duration is known only after the call has taken place. Therefore, it was excluded from the machine learning input features to avoid using information that would not be available before making the call.

5.4.5 Visualization

A bar chart was used to compare the average call duration for subscribers and non-subscribers.

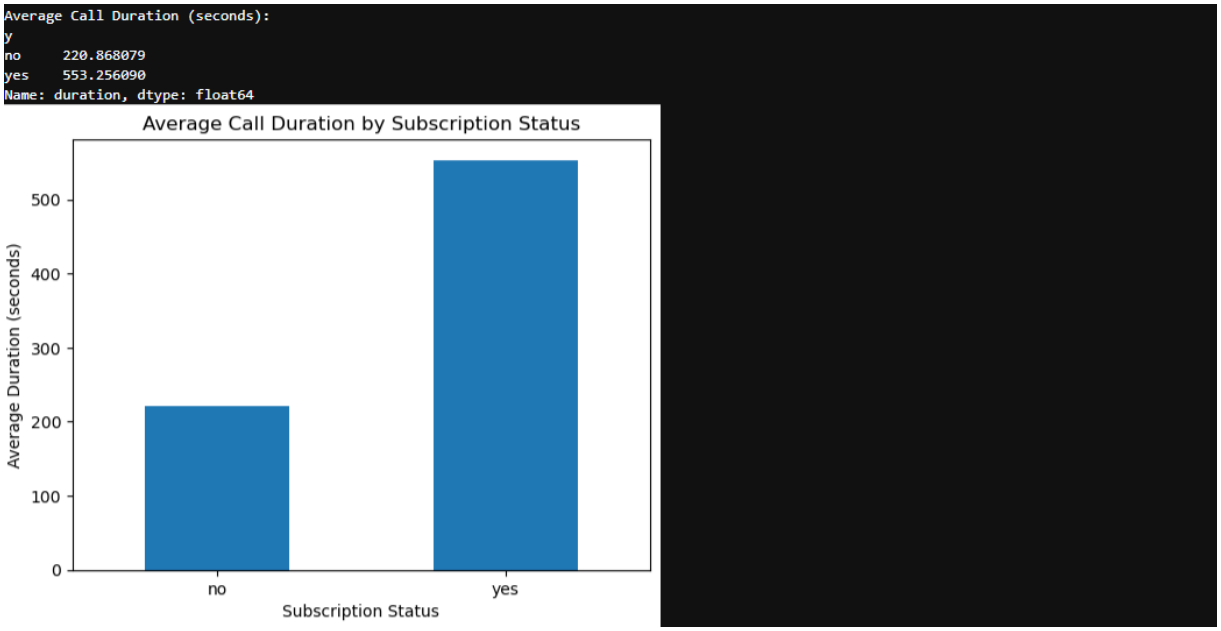


Figure 5.3: Average Call Duration by Subscription Status.

5.5 Objective 4: Compare Subscription Rates by Job and Education

5.5.1 Objective

The fourth objective was to examine how customer occupation and education level are associated with term deposit subscription behaviour.

The job and education columns were used to compare subscription rates across different customer categories.

This analysis helps understand which occupational and educational groups showed relatively higher subscription rates in the dataset.

5.5.2 Methodology

The subscription rate was calculated by grouping the dataset according to occupation and education level.

The groupby() function was used to calculate the percentage of customers who subscribed within each category.

Python code:

```
job_analysis = df.groupby("job")["y"].apply(
    lambda x: (x == "yes").mean() * 100
)


education_analysis = df.groupby("education")["y"].apply(
    lambda x: (x == "yes").mean() * 100
)
```

```
print(job_analysis.sort_values(ascending=False))
print(education_analysis.sort_values(ascending=False))
```

5.5.3 Analysis Results — Job Categories

The analysis identified the following subscription rates among selected job categories:

Job category	Subscription rate
Student	31.43%
Retired	25.26%
Unemployed	14.20%
Admin	12.97%
Management	11.22%
Blue-collar	6.90%

5.5.4 Interpretation — Job Categories

The results show that students recorded the highest subscription rate among the listed job categories, at 31.43%.

Retired customers also showed a relatively high subscription rate of 25.26%.

Blue-collar customers recorded a lower subscription rate of 6.90%.

These findings suggest that subscription behaviour differs across occupational groups.

The results should be interpreted carefully because different job categories may contain different numbers of customers. A high subscription percentage in a small group does not necessarily mean that the group contributed the largest number of subscriptions.

5.5.5 Analysis Results — Education Categories

The following subscription rates were identified for the education categories:

Education category	Subscription rate
Illiterate	22.22%
Unknown	14.51%
University degree	13.72%
Professional course	11.35%
High school	10.84%
Basic 4y	10.25%
Basic 6y	8.21%
Basic 9y	7.83%

5.5.6 Interpretation — Education Categories

The results indicate that customers in the illiterate category recorded the highest subscription rate among the listed education categories, at 22.22%.

Customers with a university degree recorded a subscription rate of 13.72%.

The results show that subscription rates varied across education categories. However, these differences should not be interpreted as evidence that education directly determines subscription decisions.

The number of customers in each category should also be considered when interpreting the results.

5.5.7 Visualization

Bar charts were used to compare subscription rates across job categories and education levels.

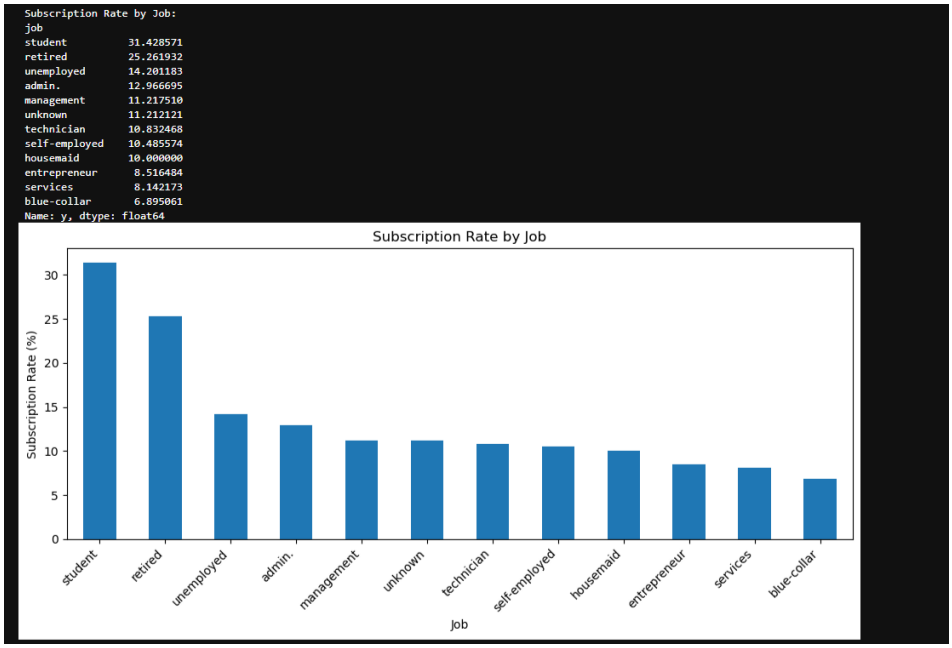


Figure 5.4: Term Deposit Subscription Rate by Job Category.

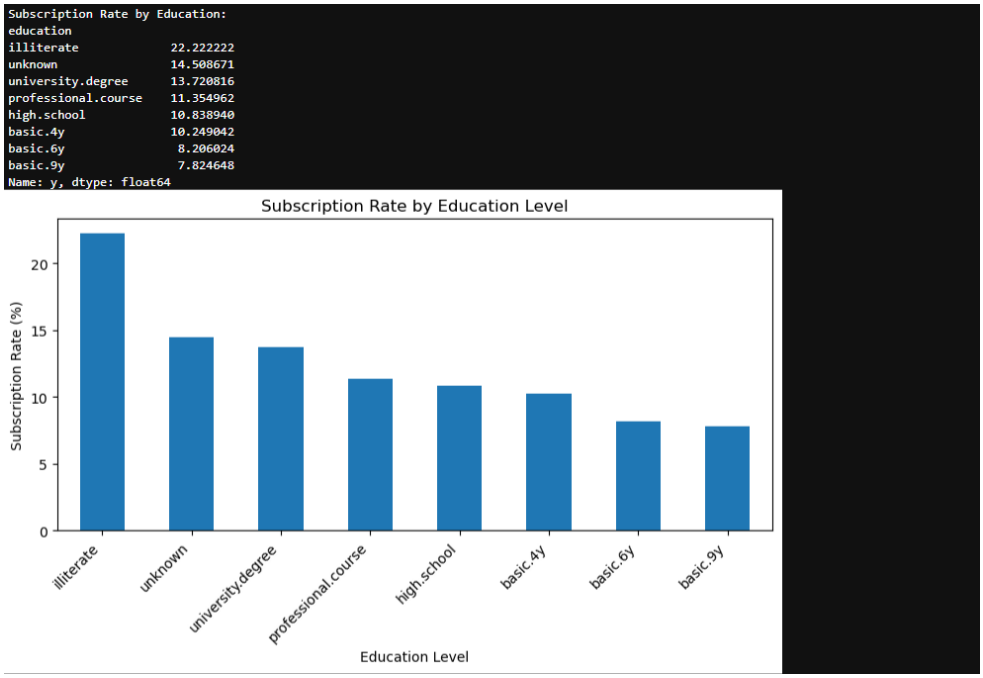


Figure 5.5: Term Deposit Subscription Rate by Education Level.

5.6 Objective 5: Analyze Previous Campaign Outcomes

5.6.1 Objective

The fifth objective was to study how the results of previous marketing campaigns are associated with subscription behaviour in the current campaign.

The poutcome column represents the outcome of the previous marketing campaign.

The analysis was performed to understand whether customers who had a successful previous campaign experience showed different subscription behaviour compared with other customers.

5.6.2 Methodology

The subscription rate was calculated by grouping the dataset according to the poutcome column.

The analysis included the following categories:

- Success
- Failure
- Nonexistent

Python code:

```
poutcome_analysis = df.groupby("poutcome")["y"].apply(
   lambda x: (x == "yes").mean() * 100
)


print(poutcome_analysis)
```

5.6.3 Analysis Results

The analysis produced the following subscription rates:

Previous campaign outcome	Subscription rate
Success	65.11%
Failure	14.23%
Nonexistent	8.83%

5.6.4 Interpretation

The results show that customers with a successful previous campaign outcome recorded the highest subscription rate of 65.11%.

Customers whose previous campaign outcome was failure recorded a subscription rate of 14.23%.

Customers with no previous campaign outcome recorded a subscription rate of 8.83%.

This indicates that previous campaign success is strongly associated with subscription behaviour in the current campaign.

The findings suggest that previous campaign information can provide useful information for understanding customer responses.

5.6.5 Visualization

A bar chart was used to compare subscription rates according to previous campaign outcomes.

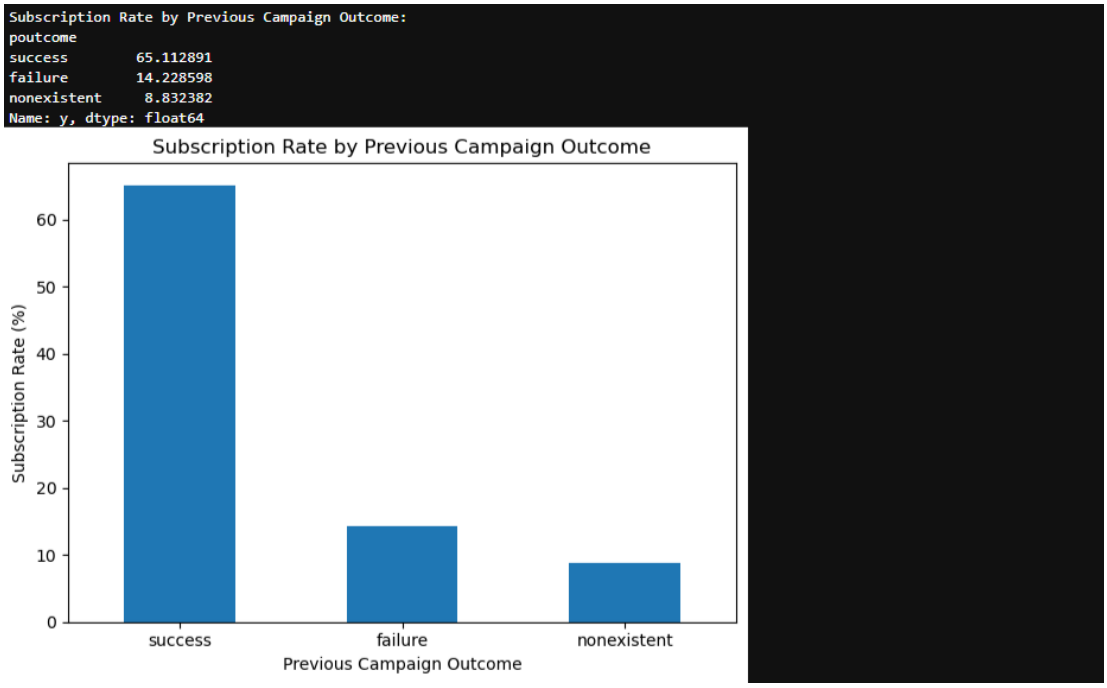


Figure 5.6: Subscription Rate by Previous Campaign Outcome.

5.7 Objective 6: Compare Subscription Rates by Contact Method

5.7.1 Objective

The sixth objective was to compare customer subscription rates according to the contact method used by the bank.

The contact column contains information about the communication method used to contact customers.

The analysis focused on comparing two contact methods:

- Cellular
- Telephone

The purpose was to understand whether subscription rates differed according to the communication channel.

5.7.2 Methodology

The subscription rate was calculated for each contact method using the groupby() function.

Python code:

```
contact_analysis = df.groupby("contact")["y"].apply(
    lambda x: (x == "yes").mean() * 100
)

print(contact_analysis)
```

5.7.3 Analysis Results

The analysis produced the following subscription rates:

Contact method	Subscription rate
Cellular	14.74%
Telephone	5.23%

5.7.4 Interpretation

The results show that customers contacted through cellular communication recorded a subscription rate of 14.74%.

Customers contacted through telephone communication recorded a subscription rate of 5.23%.

The findings indicate that the cellular contact category had a higher subscription rate in the dataset.

This difference may be related to customer preferences, campaign practices, or other factors. The analysis does not establish that the contact method alone caused the difference in subscription rates.

5.7.5 Visualization

A bar chart was used to compare subscription rates between cellular and telephone contact methods.

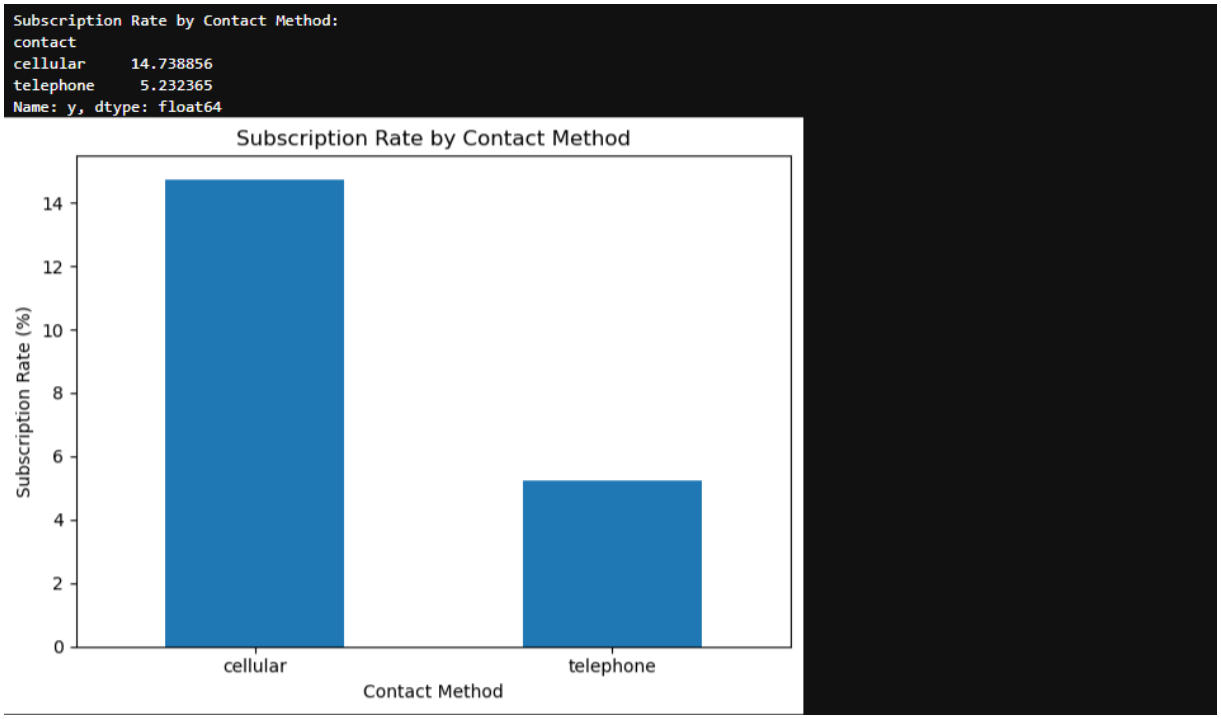


Figure 5.7: Subscription Rate by Contact Method.

5.8 Overall Summary of the Six Objectives

The six objectives provided different perspectives on customer subscription behaviour.

The major findings are summarized below:

Objective	Key finding

Overall subscription rate	11.27% subscribed
Age group analysis	18–25 had the highest rate at 20.95%
Call duration	Subscribers averaged 553.26 seconds
Job analysis	Students had the highest listed rate at 31.43%
Previous campaign outcome	Previous success had a 65.11% rate
Contact method	Cellular had a 14.74% rate

5.9 Conclusion

The analysis of the six objectives helped identify several patterns associated with customer term deposit subscriptions.

The overall subscription rate was relatively low, with only 11.27% of customers subscribing. Subscription rates also varied across age groups, occupations, education levels, previous campaign outcomes, and contact methods.

One of the strongest observed patterns was the high subscription rate among customers whose previous campaign outcome was successful. Call duration also showed a noticeable difference between subscribers and non-subscribers.

These findings helped identify potentially useful variables for machine learning classification. The next stage focuses on preparing the input features and target variable for model development.

6.FEATURE ENGINEERING AND MACHINE LEARNING DATA PREPARATION

6.1 Introduction

Feature engineering and data preparation are important stages in machine learning. Before training a classification model, the dataset must be converted into a suitable format that machine learning algorithms can understand.

The Bank Marketing dataset contains both numerical and categorical variables. Numerical variables contain measurable values, while categorical variables represent groups or labels.

In this project, the cleaned dataset was prepared by selecting input features, separating the target variable, converting categorical variables into numerical form, splitting the data into training and testing sets, and applying feature scaling.

These steps helped prepare the dataset for Logistic Regression, K-Nearest Neighbors, Decision Tree, and Gaussian Naive Bayes algorithms.

6.2 Understanding Features and Target Variable

In machine learning, a feature is an input variable used by a model to make predictions. The target variable is the output that the model attempts to predict.

In this project:

- **Input features (X):** Customer information, marketing campaign details, previous campaign information, and economic indicators.

- **Target variable (y):** The y column, which indicates whether the customer subscribed to a term deposit.

The target variable contains two categories: yes and no.

The purpose of the classification models is to predict the subscription decision using the available input features.

6.3 Feature Selection

Feature selection involves identifying the variables that will be used as inputs for machine learning models.

The dataset contains several customer and campaign-related attributes. However, some variables require special consideration before model training.

The duration column represents the duration of the last telephone call. This information is known only after the call has been completed. Therefore, it was removed from the machine learning input features because it would not be available when making a prediction before contacting the customer.

The age_group column was also removed because it was created for exploratory analysis and represents information already available in the original age column.

This helped avoid using redundant information in the model.

6.4 Separating Input and Output Variables

The input variables and target variable were separated before training the models.

The input dataset was created by removing the target column from the original DataFrame. The target variable was stored separately.

Python code:

```
ml_df = df.copy()


ml_df = ml_df.drop(
   ["duration", "age_group"],
   axis=1
)


X = ml_df.drop("y", axis=1)
y = ml_df["y"]
```

Here:

- ml_df contains the dataset prepared for machine learning.
- X contains the input features.
- y contains the target variable.

6.5 Encoding the Target Variable

Machine learning algorithms generally require the target classes to be represented numerically.

The target variable originally contains two categories: yes and no.

These categories were converted into numerical values using mapping.

Python code:

```
y = y.map({
   "no": 0,
   "yes": 1
})
```

The conversion is shown below:

Original value	Encoded value
No	0
Yes	1

The value 0 represents customers who did not subscribe, while 1 represents customers who subscribed.

6.6 Encoding Categorical Features

The Bank Marketing dataset contains categorical variables such as job, marital status, education, contact method, and previous campaign outcome.

Most machine learning algorithms require numerical input. Therefore, categorical variables must be converted into numerical representations.

In this project, One-Hot Encoding was performed using the pd.get_dummies() function.

Python code:

```
X = pd.get_dummies(
   X,
   drop_first=True,
   dtype=int
)
```

One-Hot Encoding creates separate binary columns for categorical values.

For example, a categorical variable representing contact method can be converted into separate indicator columns.

The drop_first=True parameter removes one category from each encoded group to reduce redundant columns.

The dtype=int parameter ensures that the encoded values are stored as integers.

6.7 Train-Test Splitting

After encoding the categorical variables, the dataset was divided into training and testing sets.

The training dataset is used to teach the machine learning algorithm, while the testing dataset is used to evaluate how well the trained model performs on unseen records.

An 80:20 split was used in this project.

- 80% of the records were used for training.
- 20% of the records were used for testing.

The train_test_split() function from Scikit-learn was used.

Python code:

```
from sklearn.model_selection import train_test_split


X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

The parameters used are explained below:

- test_size=0.2: Assigns 20% of the records to the testing set.
- random_state=42: Makes the splitting process reproducible.
- stratify=y: Maintains approximately the same target-class proportions in both training and testing sets.

After splitting, the training data was used to fit the models and the testing data was used for performance evaluation.

6.8 Feature Scaling

Feature scaling is a preprocessing technique used to bring numerical features onto a comparable scale.

Some machine learning algorithms, such as Logistic Regression and K-Nearest Neighbors, can be affected by differences in feature magnitudes.

For example, age may contain values in tens, while economic indicators may contain values with different numerical ranges.

StandardScaler was used to standardize the input features.

Python code:

```
from sklearn.preprocessing import StandardScaler


scaler = StandardScaler()


X_train_scaled = scaler.fit_transform(X_train)


X_test_scaled = scaler.transform(X_test)
```

The fit_transform() function calculates the scaling parameters from the training data and applies the transformation to it.

The transform() function applies the same scaling parameters to the testing data.

This approach prevents information from the testing dataset from being used while learning the scaling parameters.

6.9 Importance of Data Preparation

The data preparation stage is important because machine learning algorithms require properly formatted input data.

The main benefits of these operations include:

1. Converting categorical information into numerical format.
2. Separating input variables and the target variable.
3. Dividing data into training and testing sets.
4. Maintaining target-class proportions through stratification.
5. Scaling features for algorithms that are sensitive to feature magnitude.
6. Excluding call duration to avoid using information available only after the call.

6.10 Final Dataset Preparation Summary

The following steps were completed before training the machine learning models:

Step	Operation	Purpose
1	Feature selection	Select suitable input variables
2	Drop duration	Avoid post-call information
3	Drop age_group	Remove redundant engineered EDA column
4	Target encoding	Convert yes/no into 1/0
5	One-Hot Encoding	Convert categorical features into numerical columns
6	Train-test split	Divide data into training and testing sets
7	Standardization	Scale features for suitable algorithms

6.11 Conclusion

Feature engineering and machine learning data preparation converted the cleaned Bank Marketing dataset into a format suitable for classification.

The target variable was encoded into numerical values, categorical input variables were converted using One-Hot Encoding, and the dataset was divided into training and testing sets using an 80:20 ratio.

Feature scaling was also applied using StandardScaler. The duration variable was excluded because it is known only after the telephone call.

After completing these steps, the prepared dataset was ready for training and evaluating the four machine learning classification algorithms.

7.IMPLEMENTATION OF MACHINE LEARNING MODELS

7.1 Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed for every situation.

In this project, machine learning classification algorithms are used to predict whether a bank customer will subscribe to a term deposit. The target variable is y, which contains two possible values:

- **Yes (1):** The customer subscribed to the term deposit.
- **No (0):** The customer did not subscribe to the term deposit.

Four classification algorithms were implemented and evaluated:

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Decision Tree Classifier
4. Gaussian Naive Bayes

The models were trained using the prepared dataset and their performance was evaluated using classification metrics.

7.2 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used mainly for binary classification problems. It predicts the probability that an observation belongs to a particular class.

In this project, Logistic Regression predicts whether a customer will subscribe to a term deposit or not.

The model uses the logistic (sigmoid) function to convert predicted values into probabilities between 0 and 1.

Implementation:

```
from sklearn.linear_model import LogisticRegression


lr_model = LogisticRegression(max_iter=1000)


lr_model.fit(X_train_scaled, y_train)


lr_pred = lr_model.predict(X_test_scaled)
```

Explanation:

- LogisticRegression() creates the model.
- max_iter=1000 allows the algorithm more iterations to reach convergence.
- fit() trains the model using training data.
- predict() generates predictions for the testing data.

The scaled training and testing datasets were used because Logistic Regression can perform better when numerical features have comparable scales.

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report

# Create the model
lr = LogisticRegression(max_iter=1000)

# Train the model
lr.fit(X_train_scaled, y_train)

# Make predictions
y_pred_lr = lr.predict(X_test_scaled)

print("Logistic Regression Model")
print("Accuracy:", accuracy_score(y_test, y_pred_lr))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_lr))

print("\nClassification Report:")
print(classification_report(y_test, y_pred_lr))
```

```
Logistic Regression Model
Accuracy: 0.8980087421078193

Confusion Matrix:
[[7202  106]
 [ 734  194]]

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.99      0.94      7308
           1       0.65      0.21      0.32       928

    accuracy                           0.90      8236
   macro avg       0.78      0.60      0.63      8236
weighted avg       0.88      0.90      0.87      8236
```

7.3 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a supervised learning algorithm used for classification and regression. It predicts the class of a new observation by identifying the nearest observations in the training dataset.

In this project, KNN checks the five nearest customer records and predicts whether the customer is likely to subscribe based on their class labels.

Implementation:

from sklearn.neighbors import KNeighborsClassifier

knn_model = KNeighborsClassifier(n_neighbors=5)

knn_model.fit(X_train_scaled, y_train)

knn_pred = knn_model.predict(X_test_scaled)

Explanation:

- KNeighborsClassifier() creates the KNN classification model.
- n_neighbors=5 means the model considers five nearest neighbours.
- fit() stores the training data for classification.
- predict() predicts the subscription outcome for test records.

Feature scaling is important for KNN because it uses distance calculations. Without scaling, features with larger numerical ranges could dominate the distance calculation.

```python
from sklearn.neighbors import KNeighborsClassifier

# Create KNN model
knn = KNeighborsClassifier(n_neighbors=5)

# Train the model
knn.fit(X_train_scaled, y_train)

# Make predictions
y_pred_knn = knn.predict(X_test_scaled)

# Evaluate the model
print("KNN Model")
print("Accuracy:", accuracy_score(y_test, y_pred_knn))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_knn))

print("\nClassification Report:")
print(classification_report(y_test, y_pred_knn))
```

```
KNN Model
Accuracy: 0.8908450704225352

Confusion Matrix:
[[7083  225]
 [ 674  254]]

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.97      0.94      7308
           1       0.53      0.27      0.36       928

    accuracy                           0.89      8236
   macro avg       0.72      0.62      0.65      8236
weighted avg       0.87      0.89      0.88      8236
```

7.4 Decision Tree Classifier

A Decision Tree is a supervised machine learning algorithm that makes predictions by dividing data into smaller groups using decision rules.

It works like a flowchart. Each internal node represents a decision based on a feature, branches represent possible outcomes, and leaf nodes represent the final prediction.

In this project, the Decision Tree uses customer information to classify whether a customer will subscribe to a term deposit.

```python
from sklearn.tree import DecisionTreeClassifier

# Create Decision Tree model
dt = DecisionTreeClassifier(
    criterion="entropy",
    max_depth=8,
    random_state=42
)

# Train the model
dt.fit(X_train, y_train)

# Make predictions
y_pred_dt = dt.predict(X_test)

# Evaluate the model
print("Decision Tree Model")
print("Accuracy:", accuracy_score(y_test, y_pred_dt))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_dt))

print("\nClassification Report:")
print(classification_report(y_test, y_pred_dt))
```

```
Decision Tree Model
Accuracy: 0.898251578436134

Confusion Matrix:
[[7146  162]
 [ 676  252]]

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.98      0.94      7308
           1       0.61      0.27      0.38       928

    accuracy                           0.90      8236
   macro avg       0.76      0.62      0.66      8236
weighted avg       0.88      0.90      0.88      8236
```

Implementation:

```
from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier(
    criterion="entropy",
    max_depth=8,
    random_state=42
)

dt_model.fit(X_train, y_train)

dt_pred = dt_model.predict(X_test)
```

Explanation:

- DecisionTreeClassifier() creates the classification model.
- criterion="entropy" uses information gain to select splitting decisions.
- max_depth=8 limits the maximum depth of the tree.
- random_state=42 helps make the results reproducible.
- fit() trains the model.
- predict() generates predictions for the test dataset.

The Decision Tree was trained using encoded but unscaled features. Scaling is generally not required for Decision Tree models because their splitting decisions are based on feature thresholds rather than distance calculations.

7.5 Gaussian Naive Bayes

Gaussian Naive Bayes is a supervised classification algorithm based on Bayes' Theorem. It assumes that the features follow a Gaussian (normal) distribution within each class.

The algorithm calculates the probability of each possible class and predicts the class with the higher probability.

In this project, Gaussian Naive Bayes predicts whether a customer will subscribe to a term deposit based on the available customer features.

```python
from sklearn.naive_bayes import GaussianNB

# Create Naive Bayes model
nb = GaussianNB()

# Train the model
nb.fit(X_train_scaled, y_train)

# Make predictions
y_pred_nb = nb.predict(X_test_scaled)

# Evaluate the model
print("Naive Bayes Model")
print("Accuracy:", accuracy_score(y_test, y_pred_nb))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_nb))

print("\nClassification Report:")
print(classification_report(y_test, y_pred_nb))
```

```
Naive Bayes Model
Accuracy: 0.7344584749878582

Confusion Matrix:
[[5378 1930]
 [ 257  671]]

Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.74      0.83      7308
           1       0.26      0.72      0.38       928

    accuracy                           0.73      8236
   macro avg       0.61      0.73      0.61      8236
weighted avg       0.88      0.73      0.78      8236
```

Implementation:

from sklearn.naive_bayes import GaussianNB

nb_model = GaussianNB()

nb_model.fit(X_train_scaled, y_train)

nb_pred = nb_model.predict(X_test_scaled)

Explanation:

- GaussianNB() creates the Naive Bayes classification model.
- fit() trains the model using the training dataset.
- predict() predicts the target class for the testing dataset.

Gaussian Naive Bayes was trained using scaled features in this project.

7.6 Training and Testing Approach

The dataset was divided into training and testing subsets using an 80:20 ratio.

- **Training data (80%):** Used to train the machine learning models.
- **Testing data (20%):** Used to evaluate how well the models predict unseen records.

The stratify=y parameter was used during the train-test split to maintain approximately the same proportion of subscription and non-subscription classes in both subsets.

A fixed random state of 42 was used to make the split reproducible.

7.7 Summary of Implemented Models

Algorithm	Main technique	Scaling used
Logistic Regression	Probability-based classification	Yes
KNN	Distance-based classification	Yes
Decision Tree	Rule-based classification	No
Gaussian Naive Bayes	Bayes' Theorem	Yes

All four algorithms were implemented for the same binary classification problem. Their predictions were generated using the testing dataset. The next chapter compares their performance using accuracy, precision, recall, F1-score, and confusion matrices.

7.8 Chapter Summary

In this chapter, four machine learning classification algorithms were implemented to predict bank term deposit subscriptions. Each algorithm uses a different approach to identify patterns in customer data.

Logistic Regression uses probability-based classification, KNN uses distance between observations, Decision Tree uses decision rules, and Gaussian Naive Bayes uses probability calculations based on Bayes' Theorem.

The models were trained and used to generate predictions on the testing dataset. Their performance is discussed in the next chapter.

8.MODEL EVALUATION AND COMPARISON

8.1 Introduction

After implementing the four machine learning classification algorithms, the next step is to evaluate their performance. Model evaluation helps us understand how accurately a model predicts whether a bank customer will subscribe to a term deposit.

In this project, four models were evaluated:

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Decision Tree Classifier
4. Gaussian Naive Bayes

The models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices. These metrics help compare the performance of different classification algorithms.

8.2 Evaluation Metrics

8.2.1 Accuracy

Accuracy measures the percentage of total predictions that the model classified correctly.

Formula:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

A higher accuracy indicates that the model correctly classified a larger proportion of the testing records. However, accuracy alone may be misleading when the target classes are imbalanced.

8.2.2 Precision

Precision measures how many customers predicted as subscribers actually subscribed.

Formula:

Precision = TP / (TP + FP)

Higher precision means fewer false positive predictions.

8.2.3 Recall

Recall measures how many actual subscribers were correctly identified by the model.

Formula:

Recall = TP / (TP + FN)

Higher recall means the model identifies a larger proportion of actual subscribers.

8.2.4 F1-Score

The F1-score is the harmonic mean of precision and recall. It provides a balance between these two metrics.

Formula:

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

The F1-score is useful when the dataset contains imbalanced target classes.

8.3 Confusion Matrix

A confusion matrix is a table used to evaluate the performance of a classification model. It compares actual class values with predicted class values.

The confusion matrix contains four important values:

- **True Positive (TP):** Actual subscribers correctly predicted as subscribers.
- **True Negative (TN):** Actual non-subscribers correctly predicted as non-subscribers.
- **False Positive (FP):** Non-subscribers incorrectly predicted as subscribers.
- **False Negative (FN):** Actual subscribers incorrectly predicted as non-subscribers.

In this project, the positive class is represented by 1 (Yes), while the negative class is represented by 0 (No).

8.4 Logistic Regression Evaluation

Logistic Regression achieved an accuracy of approximately 89.80%.

Its confusion matrix was:

Actual / Predicted	No (0)	Yes (1)
No (0)	7,202	106
Yes (1)	734	194

The model correctly classified 7,202 non-subscribers and 194 subscribers. However, it failed to identify 734 actual subscribers.

The model achieved approximately 65% precision and 21% recall for the subscriber class. This indicates that its positive predictions were reasonably precise, but it missed many actual subscribers.

8.5 K-Nearest Neighbors Evaluation

The KNN model achieved an accuracy of approximately 89.08%.

Its confusion matrix was:

Actual / Predicted	No (0)	Yes (1)
No (0)	7,083	225
Yes (1)	674	254

The model correctly classified 7,083 non-subscribers and 254 subscribers. It incorrectly classified 225 non-subscribers as subscribers and missed 674 actual subscribers.

The model achieved approximately 53% precision and 27% recall for the subscriber class.

8.6 Decision Tree Evaluation

The Decision Tree model achieved an accuracy of approximately 89.83%, which was the highest accuracy among the four models.

Its confusion matrix was:

Actual / Predicted	No (0)	Yes (1)
No (0)	7,146	162
Yes (1)	676	252

The model correctly predicted 7,146 non-subscribers and 252 subscribers. It also produced 162 false positive predictions and 676 false negative predictions.

The model achieved approximately 61% precision and 27% recall for the subscriber class.

8.7 Gaussian Naive Bayes Evaluation

Gaussian Naive Bayes achieved an accuracy of approximately 73.44%.

Its confusion matrix was:

Actual / Predicted	No (0)	Yes (1)
No (0)	5,378	1,930
Yes (1)	257	671

The model correctly classified 5,378 non-subscribers and 671 subscribers. It identified more actual subscribers than the other three models.

The model achieved approximately 26% precision and 72% recall for the subscriber class. Although its accuracy was lower, it identified a larger proportion of actual subscribers.

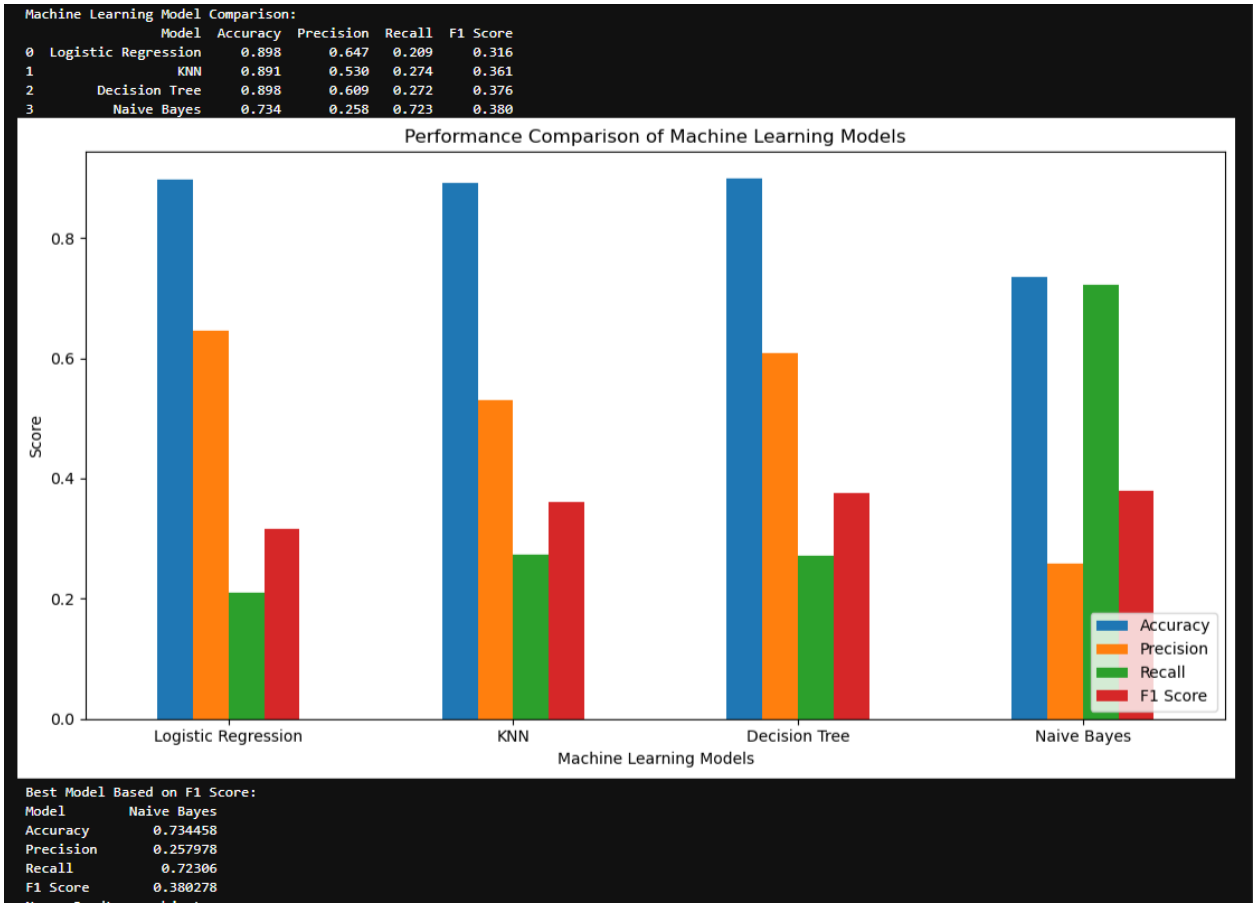
8.8 Comparison of All Four Models

The following table compares the performance of all four classification algorithms.

Model	Accuracy	Precision	Recall	F1-Score

Logistic Regression	89.80%	65%	21%	0.316
KNN	89.08%	53%	27%	0.361
Decision Tree	89.83%	61%	27%	0.376
Gaussian Naive Bayes	73.44%	26%	72%	0.380

Note: The precision, recall and F1-score values are rounded. They refer to the positive subscriber class (Yes).



8.9 Interpretation of Model Comparison

The evaluation results show that each algorithm performs differently.

- **Logistic Regression:** Achieved high accuracy and relatively high precision, but its recall was low.
- **KNN:** Achieved reasonable accuracy and identified more subscribers than Logistic Regression.
- **Decision Tree:** Achieved the highest accuracy of approximately 89.83% and a good balance between precision and recall compared with the first two models.
- **Gaussian Naive Bayes:** Achieved the highest recall and F1-score among the four models, although its accuracy and precision were lower.

The dataset contains more non-subscribers than subscribers. Therefore, accuracy alone is not sufficient to select the most suitable model.

8.10 Selection Criteria

For this project, Gaussian Naive Bayes was selected as the preferred model when considering the F1-score and recall of the subscriber class.

Its F1-score was approximately 0.380, and its recall was approximately 72%. This means it identified a larger proportion of actual subscribers than the other models.

However, the model also produced a high number of false positive predictions. Therefore, it may require further improvement before being used in a real banking environment.

The Decision Tree model achieved the highest accuracy, while Gaussian Naive Bayes provided the highest recall and F1-score.

8.11 Chapter Summary

This chapter evaluated four machine learning classification models using accuracy, precision, recall, F1-score, and confusion matrices.

The results showed that the Decision Tree achieved the highest accuracy of approximately 89.83%. Gaussian Naive Bayes achieved the highest recall of approximately 72% and the highest F1-score of approximately 0.380.

Based on the F1-score and recall, Gaussian Naive Bayes was selected as the preferred model for identifying potential term deposit subscribers. The next chapter discusses the selected model and the feature importance analysis.

9.BEST MODEL SELECTION AND FEATURE IMPORTANCE

9.1 Introduction

After evaluating four machine learning classification algorithms, the next step is to identify the most suitable model for predicting bank term deposit subscriptions.

The models were compared using accuracy, precision, recall, and F1-score. Since the dataset contains significantly more non-subscribers than subscribers, selecting a model only on the basis of accuracy may not provide the best result.

Therefore, this project considers the F1-score and recall of the subscriber class while selecting the preferred model.

9.2 Selection of the Best Model

The comparison results showed that Gaussian Naive Bayes achieved the highest F1-score and recall among the four models.

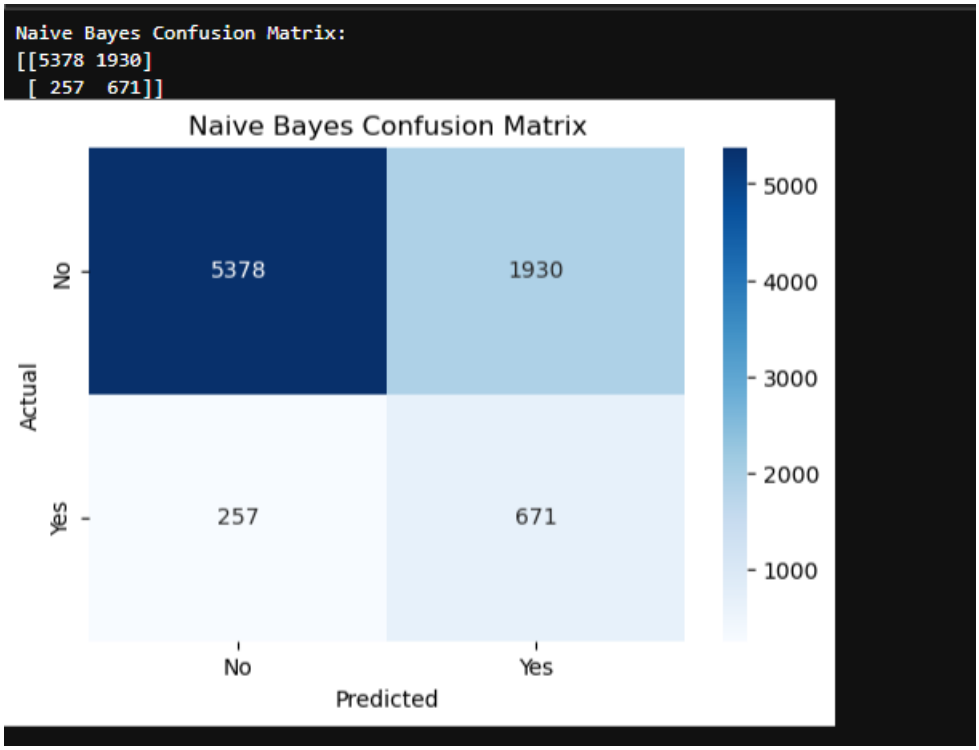
The main results were:

- Accuracy: Approximately 73.44%
- Precision: Approximately 26%
- Recall: Approximately 72%
- F1-score: Approximately 0.380

Gaussian Naive Bayes identified a larger proportion of actual subscribers than the other three models.

Although the Decision Tree achieved the highest accuracy of approximately 89.83%, its recall was considerably lower.

Based on the objective of identifying potential subscribers, Gaussian Naive Bayes was selected as the preferred model for this project.



9.3 Why Gaussian Naive Bayes Was Selected

Gaussian Naive Bayes was selected because:

1. It achieved the highest recall among the four evaluated models.
2. It achieved the highest F1-score.
3. It identified more actual subscribers than the other models.
4. It provides a useful alternative when the objective is to reduce missed subscriber predictions.

However, the model also produced a large number of false positive predictions. Therefore, its predictions should be interpreted carefully.

The selected model is preferred according to the F1-score and recall criteria, not because it achieved the highest accuracy.

9.4 Understanding Feature Importance

Feature importance is a technique used to identify which input variables contribute most to the predictions made by a machine learning model.

In this project, feature importance was calculated using the Decision Tree Classifier.

The Decision Tree assigns an importance score to each feature based on how much that feature contributes to reducing impurity across the tree's splits.

A higher importance score indicates a greater contribution to the Decision Tree's predictions.

Feature importance does not prove that a feature causes a customer to subscribe. It only indicates its predictive contribution within the trained model.

9.5 Decision Tree Feature Importance Results

The Decision Tree feature importance analysis identified the following important features:

Feature	Importance Score
nr.employed	0.575009
cons.conf.idx	0.088330*
pdays	0.079901
euribor3m	0.058727
age	0.035877
campaign	0.028191
cons.price.idx	0.027804
month_oct	0.018533
poutcome_nonexistent	0.010505
contact_telephone	0.009144

Verify the cons.conf.idx value against the final notebook output before submission.

The results indicate that nr.employed received the highest importance score among the listed features.

Other important features included consumer confidence index, previous contact information, and the Euribor three-month rate.

9.6 Interpretation of Important Features

9.6.1 Number of Employees (nr.employed)

This feature received the highest importance score in the Decision Tree model. It represents the number of employees in the economy and may capture broader economic conditions associated with customer subscription behaviour.

9.6.2 Consumer Confidence Index (cons.conf.idx)

This feature represents consumer confidence conditions. Its importance suggests that economic confidence information contributed to the Decision Tree's predictions.

9.6.3 Previous Contact Information (pdays)

The pdays variable represents the number of days since the customer was last contacted during a previous campaign. A value of 999 is used in this dataset to indicate that the customer was not previously contacted.

9.6.4 Euribor Three-Month Rate (euribor3m)

This feature represents the three-month Euribor interest rate. It provides information about the economic and financial environment during the marketing campaign.

9.6.5 Customer Age (age)

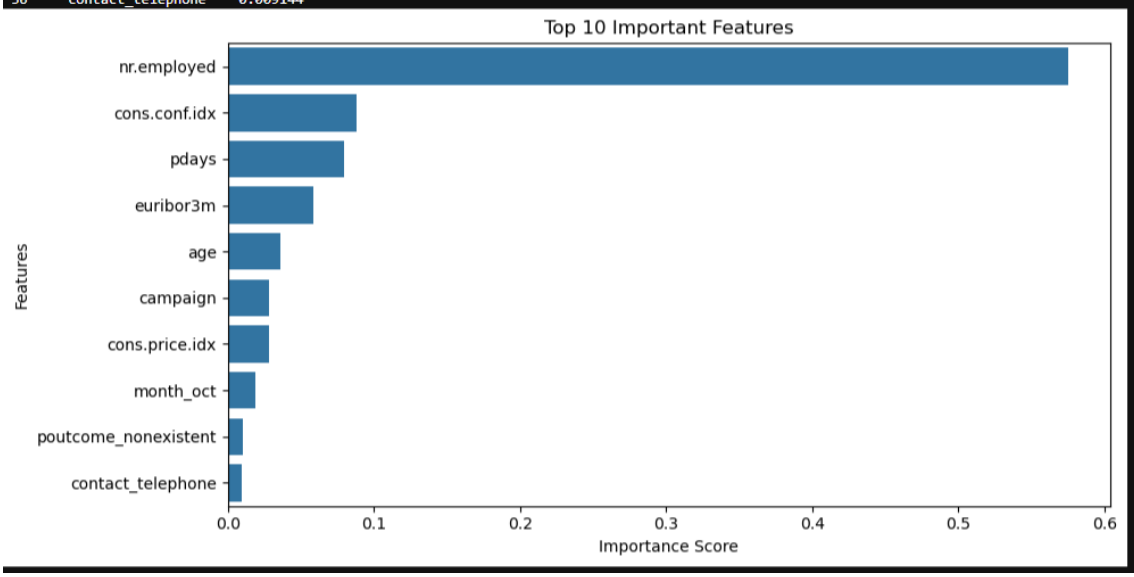
Age represents the customer's age in years. The Decision Tree model used this variable to identify patterns associated with subscription predictions.

9.6.6 Number of Contacts (campaign)

The campaign variable represents the number of contacts performed during the current campaign for a particular customer. It helps the model identify patterns related to contact frequency.

9.7 Importance of Feature Analysis

```
Top 10 Important Features:
                Feature  Importance
8          nr.employed    0.575009
6       cons.conf.idx     0.088330
2               pdays     0.079901
7           euribor3m     0.058727
0                 age     0.035877
1            campaign     0.028191
5       cons.price.idx    0.027804
44          month_oct     0.018533
50  poutcome_nonexistent  0.010505
36    contact_telephone   0.009144
```



Feature importance provides useful information about the variables that influence the predictions of the Decision Tree model.

It helps in:

- Understanding which variables are most useful for prediction.
- Identifying important economic and customer-related information.
- Interpreting the behaviour of the trained model.
- Supporting further feature selection and model improvement.

It is important to remember that the feature importance results belong to the Decision Tree model. They should not be interpreted as feature importance scores for Gaussian Naive Bayes.

9.8 Chapter Summary

This chapter discussed the selection of the preferred machine learning model and the feature importance analysis.

Gaussian Naive Bayes was selected based on its highest recall of approximately 72% and F1-score of approximately 0.380. The Decision Tree achieved the highest accuracy of approximately 89.83%.

The Decision Tree feature importance analysis identified nr.employed as the most important feature among the listed variables.

These findings provide a better understanding of the model evaluation results and the customer and economic features used for subscription prediction.

10.RESULTS AND DISCUSSION

10.1 Introduction

This chapter presents the overall results obtained from the Bank Marketing Subscription Prediction project. The project used data preprocessing, exploratory data analysis, feature engineering, and four machine learning classification algorithms to understand customer subscription behaviour.

The main purpose was to identify patterns in customer information and evaluate how effectively machine learning models could predict whether a customer would subscribe to a term deposit.

10.2 Dataset Cleaning Results

The original dataset contained 41,188 records and 21 columns.

During preprocessing, the dataset was checked for missing values and duplicate records.

The following results were obtained:

Parameter	Result
Original records	41,188
Total columns	21
Missing values	0
Duplicate records	12
Records after duplicate removal	41,176

The cleaning process removed 12 duplicate records. No missing values were found in the original dataset.

This prepared the dataset for further analysis and model development.

10.3 Exploratory Data Analysis Findings

The exploratory data analysis provided useful information about customer subscription patterns.

10.3.1 Overall Subscription Rate

The analysis showed that approximately 11.27% of customers subscribed to the term deposit, while 88.73% did not subscribe.

This indicates that the dataset contains an imbalanced target variable, with a much larger number of non-subscribers.

10.3.2 Age Group Analysis

The age group analysis showed that customers in the younger and older age categories had comparatively higher subscription rates than several middle-age categories.

The 18–25 age group had a subscription rate of approximately 20.95%, while the 56+ age group had a rate of approximately 20.69%.

10.3.3 Call Duration Analysis

The average call duration for subscribers was approximately 553.26 seconds, compared with 220.87 seconds for non-subscribers.

This indicates an association between longer call duration and subscription outcomes. However, this relationship does not establish that longer calls cause subscriptions.

10.3.4 Job and Education Analysis

The job analysis showed comparatively higher subscription rates among students and retired customers.

The education analysis also showed differences in subscription rates across education categories.

These findings suggest that customer characteristics can help describe differences in subscription behaviour.

10.3.5 Previous Campaign Outcome

Customers with a successful previous campaign outcome had a subscription rate of approximately 65.11%.

This was considerably higher than the rates observed for customers with previous unsuccessful or nonexistent campaign outcomes.

10.3.6 Contact Method Analysis

The analysis showed that customers contacted through cellular communication had a subscription rate of approximately 14.74%, compared with approximately 5.23% for telephone contact.

This indicates a difference in subscription rates between the two contact methods in the analysed dataset.

10.4 Machine Learning Results

Four classification algorithms were implemented to predict the subscription outcome.

The models were evaluated using accuracy, precision, recall, and F1-score.

The comparison results are shown below:

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	89.80%	65%	21%	0.316
KNN	89.08%	53%	27%	0.361
Decision Tree	89.83%	61%	27%	0.376
Gaussian Naive Bayes	73.44%	26%	72%	0.380

The Decision Tree achieved the highest accuracy, while Gaussian Naive Bayes achieved the highest recall and F1-score for the subscriber class.

10.5 Discussion of Model Performance

The model evaluation results demonstrate that accuracy alone is not sufficient for selecting the best model in an imbalanced classification problem.

The Decision Tree correctly classified a large proportion of total records. However, its recall for actual subscribers was approximately 27%.

Gaussian Naive Bayes achieved a higher recall of approximately 72%. This means that it identified a larger proportion of actual subscribers.

However, Gaussian Naive Bayes also produced more false positive predictions and had lower precision.

Therefore, the choice of model depends on the objective of the prediction task. In this project, identifying more potential subscribers was given importance, so Gaussian Naive Bayes was selected based on its F1-score and recall.

10.6 Feature Importance Findings

The Decision Tree feature importance analysis identified nr.employed as the most important feature among the listed variables.

Other important features included:

- cons.conf.idx
- pdays
- euribor3m
- age
- campaign

These variables contributed to the prediction decisions made by the Decision Tree model.

The results suggest that both economic indicators and customer-related information were useful for identifying patterns in the dataset. However, feature importance does not establish a causal relationship.

10.7 Major Findings of the Project

The major findings of the project are:

1. The dataset contained 41,188 records and 21 attributes before duplicate removal.
2. Twelve duplicate records were removed during preprocessing.
3. The subscription rate was approximately 11.27%.
4. Subscribers had a higher average call duration than non-subscribers.
5. Previous successful campaign outcomes were associated with higher subscription rates.
6. Cellular contact had a higher subscription rate than telephone contact.
7. Decision Tree achieved the highest accuracy of approximately 89.83%.
8. Gaussian Naive Bayes achieved the highest recall and F1-score for the subscriber class.
9. nr.employed received the highest feature importance score in the Decision Tree analysis.

10.8 Practical Implications

The findings can help banks understand customer subscription patterns and improve their marketing strategies.

Banks may use customer and campaign information to identify groups that are more likely to respond positively to marketing campaigns.

The analysis also indicates that economic indicators and previous campaign outcomes may provide useful information for customer targeting.

However, machine learning predictions should be treated as decision-support information rather than guaranteed outcomes. The model should be tested further before being used in real banking operations.

10.9 Chapter Summary

This chapter discussed the major results obtained from dataset preprocessing, exploratory data analysis, and machine learning evaluation.

The analysis identified differences in subscription rates across customer age groups, job categories, education levels, previous campaign outcomes, and contact methods.

Among the four classification algorithms, Gaussian Naive Bayes was selected based on its higher recall and F1-score, while Decision Tree achieved the highest accuracy.

Overall, the project demonstrates how data science and machine learning techniques can be used to analyse bank marketing data and predict customer subscription behaviour.

11.CONCLUSION AND FUTURE SCOPE

11.1 Introduction

The project titled **"Bank Marketing Subscription Prediction Using Machine Learning"** was developed to analyse customer information and predict whether a customer is likely to subscribe to a bank term deposit.

The project involved several stages, including dataset collection, data preprocessing, exploratory data analysis, feature engineering, machine learning model implementation, and performance evaluation.

Four classification algorithms were implemented to compare their performance and identify a suitable model for predicting customer subscription behaviour.

11.2 Conclusion

The project successfully demonstrated the use of data science and machine learning techniques for analysing bank marketing data.

The original dataset contained 41,188 records and 21 attributes. During data preprocessing, 12 duplicate records were identified and removed, resulting in 41,176 records for further analysis. No missing values were found in the original dataset.

Exploratory Data Analysis helped identify important patterns in customer subscription behaviour. The overall subscription rate was approximately 11.27%, indicating an imbalance between subscribers and non-subscribers.

The analysis also showed differences in subscription rates across age groups, job categories, education levels, previous campaign outcomes, and contact methods.

Four machine learning classification algorithms were implemented:

1. Logistic Regression
2. K-Nearest Neighbors
3. Decision Tree Classifier
4. Gaussian Naive Bayes

The models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

The Decision Tree model achieved the highest accuracy of approximately 89.83%. However, Gaussian Naive Bayes achieved the highest recall of approximately 72% and the highest F1-score of approximately 0.380 for the subscriber class.

Based on the objective of identifying more potential subscribers, Gaussian Naive Bayes was selected as the preferred model.

Overall, this project improved understanding of data preprocessing, exploratory analysis, classification algorithms, model evaluation, and the practical application of machine learning in banking.

11.3 Key Achievements

The major achievements of the project include:

- Collected and analysed a bank marketing dataset containing 41,188 records.
- Performed data cleaning and removed duplicate records.
- Conducted exploratory data analysis using statistical summaries and visualizations.
- Studied six objectives related to customer subscription behaviour.
- Applied feature engineering and categorical encoding techniques.
- Implemented four machine learning classification algorithms.
- Evaluated models using multiple classification metrics.
- Selected Gaussian Naive Bayes based on recall and F1-score.
- Identified important predictive features using Decision Tree feature importance.

11.4 Future Scope

Although the project achieved its main objectives, several improvements can be made in future work.

11.4.1 Use of Additional Machine Learning Algorithms

Future work can include additional classification algorithms such as:

- Random Forest Classifier
- Support Vector Machine (SVM)
- Gradient Boosting Classifier
- XGBoost Classifier

These algorithms can be compared with the existing models to identify whether better predictive performance can be achieved.

11.4.2 Hyperparameter Tuning

The performance of machine learning algorithms can be improved by selecting suitable parameter values.

Techniques such as GridSearchCV and RandomizedSearchCV can be used to identify better model configurations.

11.4.3 Handling Imbalanced Data

Since the dataset contains more non-subscribers than subscribers, future work can explore techniques such as:

- SMOTE (Synthetic Minority Over-sampling Technique)
- Random undersampling
- Class weights
- Threshold adjustment

These techniques may help improve the identification of potential subscribers.

11.4.4 Development of a Web Application

The selected machine learning model can be integrated into a web application using Python Flask.

A user could enter customer information, and the application could display the predicted subscription outcome.

11.4.5 Use of More Recent Data

Future research can use more recent bank marketing datasets to study changes in customer behaviour and improve the relevance of predictions.

11.4.6 Model Explainability

Explainability techniques such as SHAP and LIME can be used to understand individual predictions and provide more detailed explanations of model decisions.

11.5 Recommendations

Based on the project findings, the following recommendations can be made:

1. Banks can analyse previous campaign outcomes to understand customer response patterns.
2. Customer characteristics and economic indicators can be considered when planning marketing campaigns.
3. Different classification metrics should be considered when selecting a model for imbalanced datasets.
4. Models should be tested on independent and recent datasets before practical deployment.
5. Prediction results should support human decision-making rather than replace it completely.

11.6 Final Conclusion

This project demonstrates how machine learning can be applied to bank marketing data to understand customer behaviour and predict term deposit subscriptions.

The comparison of four classification algorithms showed that different models perform differently depending on the evaluation metric. Gaussian Naive Bayes provided the highest recall and F1-score, while Decision Tree achieved the highest accuracy.

The project provided practical experience in data science, Python programming, data visualization, feature engineering, and machine learning model evaluation.

It also provides a foundation for future improvements, including advanced classification algorithms, hyperparameter tuning, and deployment through a web application.

12.LIMITATIONS OF THE PROJECT

12.1 Introduction

Although the project successfully applies data preprocessing, exploratory data analysis, feature engineering, and machine learning techniques to predict bank marketing subscriptions, some limitations may affect the results and their practical application.

Understanding these limitations is important because machine learning models depend on the quality of the dataset, selected features, model assumptions, and evaluation methods.

12.2 Dataset Limitations

The project uses a Bank Marketing dataset containing information about customers contacted during a particular bank marketing campaign. Therefore, the dataset may not represent every bank, region, or customer group.

Customer preferences and banking behaviour can change over time. As a result, patterns identified in this dataset may not remain equally accurate when applied to newer campaigns or different banking environments.

12.3 Class Imbalance

The target variable contains two categories: Yes and No. Approximately 11.27% of customers subscribed to the bank's product, while 88.73% did not subscribe.

This difference creates a class imbalance problem. A model may achieve high accuracy by predicting the majority class more frequently, while still failing to identify a sufficient number of actual subscribers.

For this reason, accuracy alone is not enough to evaluate the models. Precision, recall, F1-score, and confusion matrices are also considered.

12.4 Model Performance Limitations

Four machine learning algorithms were implemented: Logistic Regression, K-Nearest Neighbors, Decision Tree, and Gaussian Naive Bayes.

The models produced different results. Gaussian Naive Bayes achieved the highest recall and F1-score among the tested models, but it also produced a relatively high number of false positive predictions.

The Decision Tree achieved the highest accuracy among the tested models, but its recall was lower than that of Gaussian Naive Bayes.

Therefore, model selection depends on the business objective. A bank that wants to identify more potential subscribers may prefer higher recall, while a bank that wants to reduce unnecessary customer follow-ups may place greater importance on precision.

12.5 Exclusion of Call Duration

The call duration variable showed a difference between customers who subscribed and those who did not. However, call duration is known only after the customer call has finished.

Since the main objective is to predict subscription before contacting the customer, duration was excluded from the machine learning input features.

This decision makes the prediction setup more realistic for pre-call marketing decisions. However, it also means that the model cannot use information about the length of the completed conversation.

12.6 Feature Importance Limitations

Feature importance was examined using the Decision Tree model. It helped identify which features contributed more to the model's predictions.

However, feature importance does not prove that a feature directly causes a customer to subscribe. It only describes how the selected model used the available features when making predictions.

The importance values may also change if a different algorithm, dataset, or training sample is used.

12.7 Model Validation and Hyperparameter Tuning

The dataset was divided into training and testing sets using an 80:20 split. The random state was fixed to make the split reproducible, and stratification was used to preserve the target-class distribution.

The project evaluates the models on this test split. However, a separate external dataset was not used for independent validation. The project also does not establish an exhaustive hyperparameter-tuning process for all four algorithms.

Further testing with cross-validation, additional parameter combinations, and newer data could provide a more complete assessment of model performance.

12.8 Dataset-Specific Values

Some dataset fields require careful interpretation. For example, the value 999 in the pdays column represents a customer who was not previously contacted in an earlier campaign, rather than an ordinary number of days.

If such special values are interpreted incorrectly, they may lead to misleading conclusions. The meaning of dataset-specific values should therefore be considered during preprocessing and model development.

12.9 Conclusion of Limitations

The project provides a useful demonstration of how machine learning can support bank marketing subscription prediction. However, its results should be interpreted within the limits of the selected dataset, class distribution, model performance, and validation approach.

Future improvements can include using more recent and diverse customer data, applying class-imbalance handling techniques, performing systematic hyperparameter tuning, and validating the selected model on independent datasets.

13.REFERENCES

1. Moro, S., Cortez, P., & Rita, P. (2014). *A Data-Driven Approach to Predict the Success of Bank Telemarketing*. Decision Support Systems, 62, 22–31.
2. UCI Machine Learning Repository. *Bank Marketing Dataset*. https://archive.ics.uci.edu/dataset/222/bank+marketing
3. Kaggle. *Bank Marketing Dataset*. Dataset downloaded from Kaggle. (Insert the exact Kaggle dataset link used.)
4. Pandas Documentation. https://pandas.pydata.org/docs/
5. NumPy Documentation. https://numpy.org/doc/
6. Matplotlib Documentation. https://matplotlib.org/stable/
7. Seaborn Documentation. https://seaborn.pydata.org/
8. Scikit-learn Documentation. https://scikit-learn.org/stable/
9. Jupyter Notebook Documentation. https://jupyter-notebook.readthedocs.io/